

Chapter-4

Eigenvalues_and_Eigenvectors

Exercise-4.1

1. Find the eigenvalues and eigenvectors of the following matrices

a. $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ Ans: $\lambda_1 = -1, \lambda_2 = 5, \mathbf{V}_1 = (-2a, a)^T, \mathbf{V}_2 = (b, b)^T$

b. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ Ans: $\lambda_1 = 1, \lambda_2 = 3, \mathbf{V}_1 = (-a, a)^T, \mathbf{V}_2 = (b, b)^T$

c. $\begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix}$ Ans: $\lambda_1 = 1, \lambda_2 = 3, \mathbf{V}_1 = (-2a, a)^T, \mathbf{V}_2 = (0, b)^T$

d. $\begin{pmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$ Ans: $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$
 $\mathbf{V}_1 = (0, a, 0)^T, \mathbf{V}_2 = (-b, 2b, 2b)^T, \mathbf{V}_3 = (-c, c, c)^T$

e. $\begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$ Ans: $\lambda_1 = -2, \lambda_2 = 4, \lambda_3 = 6$
 $\mathbf{V}_1 = (0, a, 0)^T, \mathbf{V}_2 = (b, 0, -b)^T, \mathbf{V}_3 = (c, 0, c)^T$

2. Solve the following system of differential equations using eigenvalue and eigenvector where $\dot{x}_1(t) = \frac{dx_1}{dt}$ and $\dot{x}_2(t) = \frac{dx_2}{dt}$.

a. $\begin{cases} \dot{x}_1(t) = -x_1(t) + 4x_2(t) \\ \dot{x}_2(t) = 3x_1(t) - 2x_2(t) \end{cases}$ with $x_1(0) = 3, x_2(0) = 4$. Ans.

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = e^{2t} \begin{pmatrix} 4 \\ 3 \end{pmatrix} + e^{-5t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

b. $\begin{cases} \dot{x}_1(t) = 5x_1(t) + x_2(t) \\ \dot{x}_2(t) = x_1(t) + 5x_2(t) \end{cases}$ with $x_1(0) = 1, x_2(0) = -3$. Ans.

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = -e^{6t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2e^{4t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

c. $\begin{cases} \dot{x}_1(t) = -2x_1 + 3x_2(t) \\ \dot{x}_2(t) = 4x_1(t) - x_2(t) \end{cases}$ with $x_1(0) = 4, x_2(0) = 3$. Ans.

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = e^{2t} \begin{pmatrix} 3 \\ 4 \end{pmatrix} - e^{5t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Exercise-4.2

State the Cayley-Hamilton theorem. Hence find the inverse of the following matrices using Cayley-Hamilton theorem and verify your result.

a. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

Ans. $A^{-1} = \frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$

b. $A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

Ans. $A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{pmatrix}$

c. $A = \begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$

Ans. $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{pmatrix}$

d. $A = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$

Ans. $A^{-1} = \begin{pmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$

e. $A = \begin{pmatrix} 1 & -1 & 1 \\ 4 & 1 & 0 \\ 8 & 1 & 1 \end{pmatrix}$

Ans. $A^{-1} = \begin{pmatrix} 1 & 2 & -1 \\ -4 & -7 & 4 \\ -4 & -9 & 5 \end{pmatrix}$

Exercise-4.3

Write the vector $\mathbf{u}$ as a linear combination of the vectors $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$, where

1. $\mathbf{u} = (5, -1, 9)$, $\mathbf{u}_1 = (1, 2, 1)$, $\mathbf{u}_2 = (2, 9, 0)$, $\mathbf{u}_3 = (3, 3, 4)$. **Ans: $(1, -1, 2)$**
2. $\mathbf{u} = (2, -1, 3)$, $\mathbf{u}_1 = (1, 0, 0)$, $\mathbf{u}_2 = (2, 2, 0)$, $\mathbf{u}_3 = (3, 3, 3)$. **Ans: $(3, -2, 1)$**
3. $\mathbf{u} = (4, 2, 1, 0)$, $\mathbf{u}_1 = (6, -1, 2, 1)$, $\mathbf{u}_2 = (1, 7, -3, -2)$, $\mathbf{u}_3 = (3, 1, 0, 0)$,
 $\mathbf{u}_4 = (3, 3, -2, -1)$. **Ans: $(2, 1, -3, 0)$**

Test whether the following vectors are linearly independent or dependent.

4. $\mathbf{u}_1 = (1, -2, 3)$, $\mathbf{u}_2 = (5, 6, -1)$, $\mathbf{u}_3 = (3, 2, 1)$. **Ans: Linearly dependent.**
5. $\mathbf{u}_1 = (1, 2, 2, -1)$, $\mathbf{u}_2 = (4, 9, 9, -4)$, $\mathbf{u}_3 = (5, 8, 9, -5)$. **Ans: Linearly independent.**
6. $\mathbf{u}_1 = (-3, 0, 4)$, $\mathbf{u}_2 = (5, -1, 2)$, $\mathbf{u}_3 = (1, 1, 3)$. **Ans: Linearly independent.**
7. $\mathbf{u}_1 = (2, 2, 1)$, $\mathbf{u}_2 = (-1, -1, 0)$, $\mathbf{u}_3 = (3, 3, 4)$. **Ans: Linearly dependent.**
8. $\mathbf{u}_1 = (-2, 0, 1)$, $\mathbf{u}_2 = (3, 2, 5)$, $\mathbf{u}_3 = (6, -1, 1)$, $\mathbf{u}_4 = (7, 0, -2)$.
Ans: Linearly dependent.

Chapter-5

Coordinate systems

Exercise-5.1

1. Convert the coordinates of the following points from Cartesian to cylindrical and spherical coordinates:

a. $p_1 = (-\sqrt{2}, \sqrt{2}, 2)$

b. $p_2 = (0, 2, 2\sqrt{3})$

c. $p_3 = (4\sqrt{3}, 4, -8)$

2. Convert the coordinates of the following points from cylindrical to Cartesian and spherical coordinates:

a. $p_1 = (3, \frac{5\pi}{3}, 3)$

b. $p_2 = (4, \frac{\pi}{4}, 0)$

c. $p_3 = (2\sqrt{3}, \frac{\pi}{6}, -2)$

3. Convert the coordinates of the following points from spherical to Cartesian and cylindrical coordinates:

a. $p_1 = (\sqrt{2}, \frac{3\pi}{4}, \pi)$

b. $p_2 = (2, \frac{\pi}{4}, \frac{3\pi}{4})$

c. $p_3 = (1, \frac{\pi}{4}, \frac{\pi}{2})$

4. Transform the following vectors into cylindrical coordinates at the indicated points:

a. $A = \hat{x}(x+y) + \hat{y}(x+y) + \hat{z}z$ at $p = (4, 0, -4)$ Ans: $A = 4\hat{r} + 4\hat{\phi} - 4\hat{z}$.

b. $B = \hat{R}\sin\theta + \hat{\theta}\cos\theta + \hat{\phi}\cos^2\phi$ at $p = (2, \frac{\pi}{2}, \frac{\pi}{4})$ Ans: $B = \hat{r} + \frac{1}{2}\hat{\phi}$.

5. Transform the following vectors into spherical coordinates at the indicated points:

a. $A = \hat{x}y - \hat{y}x$ at $p = (1, -1, 0)$ Ans: $A = -\sqrt{2}\hat{\phi}$.

b. $B = \hat{z}\sin\phi$ at $p = (2, \frac{\pi}{4}, 2)$ Ans: $B = \frac{1}{2}\hat{R} - \frac{1}{2}\hat{\theta}$.

6. Transform the following vectors into cartesian coordinates at the indicated points:

a. $A = -\hat{\phi}R\sin\theta$ at $p = (1, \frac{\pi}{2}, 0)$ Ans: $A = -\hat{y}$.

b. $B = -\hat{r}\cos\phi - \hat{\phi}\sin\phi + \hat{z}z$ at $p = (2, \frac{2\pi}{3}, 2\sqrt{3})$ Ans: $B = \frac{1}{2}\hat{x} + \frac{\sqrt{3}}{2}\hat{y} + 2\sqrt{3}\hat{z}$.

Chapter - 6

Gradient, Divergence and Curl

Exercise - 6.1

1. Find the gradient of the following scalar functions at the indicated point:

(a) $T(x, y, z) = 4x^2 e^{-z} + y^3$ at the point $(1, -1, 0)$.

(b) $T(r, \phi, z) = r^2 \cos^2 \phi$, at the point $\left(2, \frac{5\pi}{3}, 2\right)$.

(c) $T(R, \theta, \phi) = \frac{\cos 2\theta}{R}$, at the point $(2, 0, \pi)$.

2. (a) Find the directional derivative (D. D.) of $T(x, y, z) = xy^2 + yz^3$ at the point $(2, 1, 1)$ in the direction $\mathbf{d} = 2\hat{x} - \hat{y} + 2\hat{z}$.

(b) Find the directional derivative (D. D.) of $T(r, \phi, z) = r^2 \sin 2\phi$ at the point $\left(2, \frac{\pi}{6}, 1\right)$ in the direction $\hat{r}$.

(c) Find the directional derivative (D. D.) of $T(R, \theta, \phi) = R^2 \cos \theta \sin \phi$ at the point $\left(2, \frac{\pi}{6}, \frac{5\pi}{3}\right)$ in the direction $\hat{R} - 2\hat{\theta}$.

3. Determine divergence and curl. Also check whether each of the following vector fields solenoidal, conservative or both.

(a) $\mathbf{A} = \hat{x}e^{5x} + \hat{y} 2 \cos(y) + \hat{z} 2 \sin z$

(b) $\mathbf{A} = \hat{r} r \sin \phi + \hat{\phi} r^2 \cos \phi + \hat{z} r e^{-5z}$

(c) $\mathbf{A} = \hat{R} \frac{5}{R^2} \sin \theta + \hat{\theta} r \cot \theta + \hat{\phi} R \sin \theta \cos \phi$

4. (a) Test whether $\mathbf{A} = \hat{x}(z^3 + 2xy) + \hat{y}x^2 + \hat{z}3xz^2$ is a conservative force field. If conservative, find the scalar potential T such that $\mathbf{A} = \nabla T$. Hence find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$.

(b) Test whether $\mathbf{A} = \hat{r}(3r^2z + \sin \phi) + \hat{\phi}(r \cos \phi) + \hat{z}(r^3)$ is a conservative force field. If conservative, find the scalar potential T such that $\mathbf{A} = \nabla T$. Hence find the work done in moving an object in this field from $(1, 0, -1)$ to $(1, \pi, 0)$.

(c) Test whether $\mathbf{A} = \hat{R}(\cos \theta) + \hat{\theta}(-\sin \theta)$ is a conservative force field. If conservative, find the scalar potential T such that $\mathbf{A} = \nabla T$. Hence find the work done in moving an object in this field from $\left(0, \frac{\pi}{4}, \frac{\pi}{3}\right)$ to $\left(1, \frac{\pi}{2}, 0\right)$.

5. Find the Laplacian of the following scalar functions:

(a) $T = 2x^2y^2z^2$.

(b) $T = \frac{4}{y^2 + z^2}$.

(c) $T = 3e^{-r} \cos \phi$.

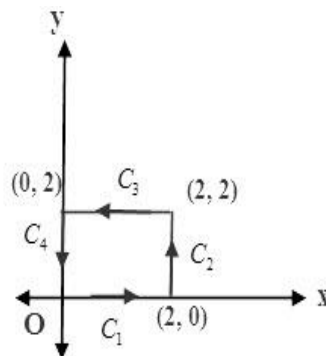
(d) $T = e^{-R} \sin \theta \cos \phi$.

Exercise - 6.2

1. For the vector field $\mathbf{A} = \hat{x} 4xz - \hat{y} y^2 + \hat{z} yz$, verify the divergence theorem by computing **(a)** the total outward flux flowing through the surface of a cube centered at the origin and with sides equal to 1 units each and parallel to the Cartesian axes, **(b)** the integral of $\nabla \cdot \mathbf{A}$ over the cube's volume.
2. For a vector function $\mathbf{A} = \hat{r} r^2 + \hat{z} 3z$, verify the divergence theorem for the circular cylindrical region enclosed by $r = 1, z = 0, z = 4$. **Ans:** 20π
3. A vector field $\mathbf{A} = \hat{r} r^2$ exists in the region between two concentric cylindrical surfaces defined by $r = 2$ and $r = 3$, with both cylinders extending between $z = 0$ and $z = 3$. Verify the divergence theorem by evaluating the following: (a) $\oint_s \mathbf{A} \cdot d\mathbf{s}$ and (b) $\int_v \nabla \cdot \mathbf{A} dv$. **Ans:** 114π
4. Find $\oint_s \mathbf{A} \cdot d\mathbf{s}$ over the surface of a hemispherical region that is the top half of a sphere of radius 4 centered at $(0, 0, 0)$ with its flat base coinciding with the xy plane. Also verify divergence theorem. where $\mathbf{A} = \hat{R} R \cos \theta$. **Ans:** 64π
5. For the vector field $\mathbf{A} = \hat{R} 2R^3$, evaluate both sides of the divergence theorem for the region enclosed between spherical shells defined by $R = 2$ and $R = 3$. **Ans:** 1688π

Exercise – 6.3

1. Verify Stokes's theorem for the vector field $\mathbf{A} = \hat{\phi} 3\sin \frac{\phi}{2}$ by evaluating it on the hemisphere of radius 2. **Ans:** 24
2. For vector field $\mathbf{A} = \hat{z} \cos \varphi$, verify Stokes's theorem for a segment of a cylindrical surface defined by $r = 5, \pi/4 \leq \varphi \leq \pi/2$, and $0 \leq z \leq 4$. **Ans:** $-2\sqrt{2}$
3. Assume that a vector field $\mathbf{A} = \hat{x}(x^2 - y^2) + \hat{y}(x^2 - xy)$, (a) find $\oint_c \mathbf{A} \cdot d\mathbf{l}$ around the rectangular contour, (b) find $\int_s (\nabla \times \mathbf{A}) \cdot d\mathbf{s}$ over rectangular arc, (c) verify Stokes's theorem and (d) can $\mathbf{A}$ be



expressed as gradient of a scalar? Explain. **Ans:** 12

4. Assume that a vector field, $\mathbf{A} = \hat{r} r \sin \phi + \hat{\phi} \cos \phi$, (a) find $\oint_C \mathbf{A} \cdot d\mathbf{l}$ over the semicircular contour shown below, and (b) find $\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s}$ over the surface of the semicircles. **Ans: 0**

